

1D Diffusion self similar solution

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1 Adding global multiplier χ

Notice that when $\chi \rightarrow \infty$, the problem will return "black" - i.e. there is a strong correlation between the radiation and the matter, and the system achieves equilibrium as $U \approx E$. It also means that the matter is a perfect absorber and emitter. The equations for $E(x, \tau)$, $U(x, \tau)$ with a global multiplier are

$$\frac{\partial E(x, \tau)}{\partial \tau} = \frac{\partial^2 E(x, \tau)}{\partial x^2} + \chi(U(x, \tau) - E(x, \tau)), \quad (1)$$

$$\frac{\partial U(x, \tau)}{\partial \tau} = \chi(E(x, \tau) - U(x, \tau)) \quad (2)$$

Using implicit finite differencing, one gets:

$$\frac{E_i^{n+1} - E_i^n}{\Delta \tau} = \frac{E_{i+1}^{n+1} - 2E_i^{n+1} + E_{i-1}^{n+1}}{(\Delta x)^2} + \chi(U_i^{n+1} - E_i^{n+1}), \quad (3)$$

$$\frac{U_i^{n+1} - U_i^n}{\Delta \tau} = \chi(E_i^{n+1} - U_i^{n+1}), \quad (4)$$

The second equation is local - solve for U_i^{n+1} :

$$(1 + \chi \Delta \tau)U_i^{n+1} - \chi \Delta \tau E_i^{n+1} = U_i^n \quad (5)$$

$$U_i^{n+1} = \frac{\chi \Delta \tau E_i^{n+1} + U_i^n}{1 + \chi \Delta \tau} \quad (6)$$

Now define $\alpha = \frac{\chi \Delta \tau}{1 + \chi \Delta \tau}$, $\lambda = \frac{\Delta \tau}{(\Delta x)^2}$ and substitute into the first equation, reaching the exact same equation as for the case $\chi = 1$ explained in `1D Diffusion Linear.tex` file.

$$(1 + 2\lambda + \alpha)E_i^{n+1} - \lambda E_{i+1}^{n+1} - \lambda E_{i-1}^{n+1} = E_i^n + \alpha U_i^n \quad (7)$$

and from here on, the solution is exactly the same as in the previous file.

2 Self-Similar Equation

Now we are going to *neglect* the assumption $C_V \sim T^3$ that makes the equation linear. Define $T = T_m$ and $\boxed{U_R(T) = aT^4}$. We also denote $\beta = \frac{\partial U_r}{\partial U_m}$ the equations now take the form:

$$\frac{\partial E}{\partial t} + \frac{\partial F}{\partial z} = \chi c \sigma (U_R - E) \quad (8)$$

$$\frac{\partial U_m}{\partial t} = -\chi c \sigma (U_R - E) \Rightarrow \frac{1}{\beta} \frac{\partial U_R}{\partial t} = -\chi c \sigma (U_R - E) \quad (9)$$

$$F = -D \frac{\partial E}{\partial z} \quad \left(D = \frac{c}{3\sigma(T)} \right) \quad (10)$$

The unknowns here are $E(z, t)$, $U_R(z, t)$ and the coefficients $D(T)$, $\beta(T)$ depend on the temperature, which is a function of U_R . In addition, U_m is also a function of T so it is also solvable from U_R (and we will use that in the simulation).

3 Self-similar relations and unit analysis

3.1 Basic relations

We assume the following self-similar relations for the opacity and the heat capacity. Equations (2a),(2b) in the Shussman-Heizler paper:

$$\frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda} \quad (11)$$

$$e = f T^\gamma \rho^{-\mu} \quad (12)$$

where e is the specific internal energy, which means the material energy density is:

$$U_m(T) = \rho e = f T^\gamma \rho^{-\mu+1} \quad (13)$$

3.2 Relations for $D(T)$, $\beta(T)$

Using the definitions of D , β and the self-similar relations for κ_R , e , we can find the relations for $D(T)$, $\beta(T)$.

$$D(T) = \frac{c}{3\sigma} = \frac{c}{3} g T^\alpha \rho^{-\lambda-1} \quad (14)$$

$$\beta(T) = \frac{\partial U_R}{\partial U_m} = \frac{4aT^3}{f\gamma T^{\gamma-1} \rho^{-\mu+1}} = \frac{4a\rho^{\mu-1}}{f\gamma} T^{4-\gamma} \quad (15)$$

from $U_m(T)$, $D(T)$ and $\beta(T)$, the simulation would be able to calculate the coefficients at each time step (see subsection 4.1).

3.3 Unit analysis and Given Constants Values

3.4 Initial and Boundary Conditions

Parameter	Symbol	Value
Initial material temperature (Kelvin)	$T_{\text{material},0}^{\text{K}}$	300 K
Boundary bath temperature (Kelvin)	$T_{\text{bath}}^{\text{K}}$	10^6 K
Initial material temperature (HeV)	$T_{\text{material},0}$	2.59×10^{-4} HeV
Boundary bath temperature (HeV)	T_{bath}	0.862 HeV
Speed of light	c	30 cm/ns
Global multiplier	χ	10^5

3.5 Simulation Grid and Time Parameters

Parameter	Symbol	Value
Domain length	L	10^{-4} cm
Number of spatial grid points	N_z	1000
Spatial step size	Δz	10^{-7} cm
Final simulation time	t_{final}	5.0 ns
Time step	Δt	5.0×10^{-4} ns
Number of time steps	N_t	10001

3.6 Physical Constants and Unit Conversions

Constant	Symbol	Value
Electron volt to Joule	eV [SI]	1.60218×10^{-19} J
Erg per Joule	erg/J	10^7 erg/J
Electron volt (erg units)	eV [cgs]	1.60218×10^{-12} erg
Hundred electron volt	HeV [cgs]	1.60218×10^{-10} erg
Boltzmann constant (Joule)	k_B [SI]	1.38065×10^{-23} J/K
Boltzmann constant (erg)	k_B [cgs]	1.3805×10^{-16} erg/K
Kelvin per HeV	K/HeV	1.1605×10^6 K/HeV
Radiation constant (Kelvin)	$a[\text{SI}]$	7.5646×10^{-15} erg/(cm ³ K ⁴)
Radiation constant (HeV)	$a[\text{cgs}]$	1.377×10^{11} erg/(cm ³ HeV ⁴)

3.7 Gold EOS Self-Similar Parameters

Parameter	Symbol	Value	Description
Specific energy coefficient	f	3.4×10^{13}	[erg/g]
Opacity coefficient	g	1.389×10^{-4}	1/7200
Opacity exponent	α	1.5	–
Specific energy exponent	γ	1.6	–
Opacity density exponent	λ	0.2	–
Specific energy density exponent	μ	0.14	–
Material density	ρ	19.32	g/cm ³

4 Preparations For the Finite Difference Scheme

4.1 Boundary and Initial Conditions

Initial Conditions: The initial temperature is $T_m = T = T_{material-0}$ everywhere, which means:

$$(T_R)_i^0 = T_i^0 = T_{material-0} \Rightarrow E_i^0 = (U_R)_i^0 = aT_{material-0}^4 \quad (16)$$

and $(U_m)_i^0$ is calculated from $T_{material-0}$ using the self-similar relation for $U_m(T)$. This implements the function `init_state`.

```

1 import numpy as np
2
3 def init_state():
4     """Initialize E, UR, Um based on T_material_0."""
5     EO = a * T_material_0**4 * np.ones(Nz)
6     URO = a * T_material_0**4 * np.ones(Nz)
7     Um0 = U_m_of_T(URO)
8     return EO, URO, Um0

```

Listing 1: initialization of the simulation state

Boundary Conditions: The boundary condition at $i = 0$ is self-similar:

$$T_0^n = T_{bath} t^\tau \Rightarrow E_0^n = aT_{bath}^4 t^{4\tau} \quad (17)$$

and the boundary condition at $i = N - 1$ is vacuum:

$$E_{N-1}^n = 0 \quad (18)$$

and this is implemented inside the function `implicit_step_self_similar_model`.

```

1 import numpy as np
2
3 def init_state():

```

```

4      """Initializing E_left and E_right based on self-similar
      boundary conditions."""
5      # Boundary conditions
6      E_left = a * (T_bath * (t**tau)) ** 4
7      E_right = 0.0
8      # ... inner cells
9      E_new[-1] = E_right
10     E_new[1:-1] = E_inner

```

Listing 2: initialization of the simulation state

4.2 Known parameters at the n-step

As seen before, the equations are implicit in E^{n+1}, U_R^{n+1} , but the coefficients $D(T), \beta(T)$ depend on the temperature at the previous time step n , which is known from the $(U_R)_i^n$ values from the previous time step! (For $n = 0$ it is known from the initial conditions, and for any other n - by induction). First, one should find the temperature at the previous time step:

$$T_i^n = \sqrt[4]{\frac{(U_R)_i^n}{a}} \quad (19)$$

and then substitute it into the relations for $D(T), \beta(T)$:

$$D_i^n = \frac{c}{3} g (T_i^n)^\alpha \rho^{-\lambda-1} = \frac{c}{3} g \left[\frac{(U_R)_i^n}{4} \right]^{\frac{\alpha}{4}} \rho^{-\lambda-1} \quad (20)$$

$$\beta_i^n = \frac{4a\rho^{\mu-1}}{f\gamma} (T_i^n)^{4-\gamma} = \frac{4a\rho^{\mu-1}}{f\gamma} \left[\frac{(U_R)_i^n}{a} \right]^{\frac{4-\gamma}{4}} \quad (21)$$

Note: The equations for D_i^n, β_i^n are known at each time step!

4.3 Caveat about the Diffusion Coefficients

Equation (10) in a discrete form will get the following form:

$$F_{i+\frac{1}{2}}^{n+1} = -D_{i+\frac{1}{2}}^n \frac{E_{i+1}^{n+1} - E_i^{n+1}}{\Delta x} \quad (22)$$

One can notice, that it is convenient to define the fluxes at the edges of the cells, while the energy density is defined at the center of the cell. The diffusion coefficient at the edge of the cell is defined as:

$$D_{i+\frac{1}{2}}^n = \frac{2D_i^n D_{i+1}^n}{D_i^n + D_{i+1}^n} \quad (23)$$

and notice that it is calculated from the previous time step n .

4.4 Adaptive Time Stepping Based on Relative Field Changes

To improve computational efficiency while maintaining numerical accuracy, the simulation employs an adaptive time-stepping scheme based on the maximum relative change of the evolving fields between consecutive time levels. Rather than using a fixed time step, the step size is adjusted dynamically to ensure that no field varies too rapidly within a single update.

Let $E(z, t)$ denote the radiation energy density and $U_R(z, t) = aT_m^4$. After advancing from time level t^n to t^{n+1} , we evaluate the maximum relative changes

$$\delta_E = \max_i \frac{|E_i^{n+1} - E_i^n|}{|E_i^{n+1}| + E_{\min}}, \quad \delta_U = \max_i \frac{|U_{R,i}^{n+1} - U_{R,i}^n|}{|U_{R,i}^{n+1}| + U_{\min}}, \quad (24)$$

where the small regularization terms

$$E_{\min} = 10^{-3} \max_i |E_i^{n+1}|, \quad U_{\min} = 10^{-3} \max_i |U_{R,i}^{n+1}|$$

prevent division by very small values near the cold region.

A target fractional change per time step, denoted dt_{fac} (typically $\text{dt}_{\text{fac}} = 0.05$), is enforced by proposing

$$\Delta t_E^* = \Delta t_n \frac{\text{dt}_{\text{fac}}}{\delta_E}, \quad \Delta t_U^* = \Delta t_n \frac{\text{dt}_{\text{fac}}}{\delta_U}. \quad (25)$$

The next time step is chosen conservatively as

$$\Delta t_{n+1} = \min(\Delta t_E^*, \Delta t_U^*, \gamma \Delta t_n, \Delta t_{\max}), \quad \Delta t_{n+1} \geq \Delta t_{\min}, \quad (26)$$

where $\gamma \approx 1.1$ limits the rate at which the time step may grow between iterations, and $\Delta t_{\min}$ and $\Delta t_{\max}$ impose absolute bounds for stability and efficiency.

Finally, when diagnostic outputs are required at prescribed times, the adaptive time step is shortened so that the numerical solution lands exactly on the requested output time. This ensures consistent temporal sampling without sacrificing the benefits of adaptive stepping.

5 Applying Finite Difference Scheme

5.1 Derivation of tridiagonal form for E_i^{n+1}

The balance equation for the energy density in cell i is:

$$\frac{E_i^{n+1} - E_i^n}{\Delta t} + \frac{F_{i+\frac{1}{2}}^n - F_{i-\frac{1}{2}}^n}{\Delta z} = \chi c \sigma (U_{R,i}^{n+1} - E_i^{n+1}) \quad (27)$$

The discretized equation for the material energy density is:

$$\frac{1}{\beta_i^n} \frac{U_{R,i}^{n+1} - U_{R,i}^n}{\Delta t} = \chi c \sigma (E_i^{n+1} - U_{R,i}^{n+1}) \quad (28)$$

and as in the $\chi = 1$ case, we solve for $U_{R_i}^{n+1}$:

$$(1 + \beta_i^n \Delta t \chi c \sigma) U_{R_i}^{n+1} = \beta_i^n \Delta t \chi c \sigma E_i^{n+1} + U_{R_i}^n \quad (29)$$

$$U_{R_i}^{n+1} = \frac{\beta_i^n \Delta t \chi c \sigma E_i^{n+1} + U_{R_i}^n}{1 + \beta_i^n \Delta t \chi c \sigma} \quad (30)$$

Substituting $U_{R_i}^{n+1}$ into the balance equation gives:

$$\frac{E_i^{n+1} - E_i^n}{\Delta t} + \frac{F_{i+\frac{1}{2}}^{n+1} - F_{i-\frac{1}{2}}^{n+1}}{\Delta z} = \chi c \sigma_i^n \left(\frac{\beta_i^n \Delta t \chi c \sigma E_i^{n+1} + U_{R_i}^n}{1 + \beta_i^n \Delta t \chi c \sigma} - E_i^{n+1} \right) \quad (31)$$

Substituting the fluxes into the balance equation gives:

$$\frac{E_i^{n+1} - E_i^n}{\Delta t} - \frac{1}{\Delta z} \left(\frac{2D_i^n D_{i+1}^n}{D_i^n + D_{i+1}^n} \frac{E_{i+1}^n - E_i^{n+1}}{\Delta z} - \frac{2D_{i-1}^n D_i^n}{D_{i-1}^n + D_i^n} \frac{E_i^{n+1} - E_{i-1}^{n+1}}{\Delta z} \right) = \quad (32)$$

$$\chi c \sigma_i^n \left(\frac{\beta_i^n \Delta t \chi c \sigma E_i^{n+1} + U_{R_i}^n}{1 + \beta_i^n \Delta t \chi c \sigma} - E_i^{n+1} \right) \quad (33)$$

In this step, we should use the self-similar relations for D_*, β_*

$$(T_m)_i^n = \sqrt[4]{\frac{(U_R)_i^n}{a}} \quad (34)$$

$$D_i^n = \frac{c}{3} g [(T_m)_i^n]^\alpha \rho^{-\lambda-1} = \frac{c}{3} g \left[\frac{(U_R)_i^n}{4} \right]^{\frac{\alpha}{4}} \rho^{-\lambda-1} \quad (35)$$

Now, after a very painful algebra, one can achieve:

$$a_i^n E_{i-1}^{n+1} + b_i^n E_i^{n+1} + c_n^i E_{i+1}^{n+1} = d_n^i \quad (36)$$

for some coefficients $a_i^n, b_i^n, c_n^i, d_n^i$. In the next subsection, we derive these coefficients step by step.

5.2 Derivation of tridiagonal form coefficients

Add definition for Flek & Cummings coefficient f here.

Step 1: Define harmonic-face diffusion coefficients. Introduce the standard harmonic averages at cell faces:

$$D_{i+\frac{1}{2}}^n \equiv \frac{2D_i^n D_{i+1}^n}{D_i^n + D_{i+1}^n}, \quad D_{i-\frac{1}{2}}^n \equiv \frac{2D_{i-1}^n D_i^n}{D_{i-1}^n + D_i^n}. \quad (37)$$

With these, the diffusion term becomes

$$-\frac{1}{\Delta z} \left(D_{i+\frac{1}{2}}^n \frac{E_{i+1}^{n+1} - E_i^{n+1}}{\Delta z} - D_{i-\frac{1}{2}}^n \frac{E_i^{n+1} - E_{i-1}^{n+1}}{\Delta z} \right). \quad (38)$$

Step 2: Simplify the matter–radiation coupling term. Define

$$A_i^n \equiv \beta_i^n \Delta t \chi c \sigma_i^n. \quad (39)$$

Then the right-hand side becomes

$$\begin{aligned} \chi c \sigma_i^n \left(\frac{A_i^n E_i^{n+1} + U_{R_i}^n}{1 + A_i^n} - E_i^{n+1} \right) &= \chi c \sigma_i^n \left(\frac{A_i^n E_i^{n+1} + U_{R_i}^n - (1 + A_i^n) E_i^{n+1}}{1 + A_i^n} \right) \\ &= \chi c \sigma_i^n \left(\frac{U_{R_i}^n - E_i^{n+1}}{1 + A_i^n} \right) \\ &= \frac{\chi c \sigma_i^n}{1 + A_i^n} U_{R_i}^n - \frac{\chi c \sigma_i^n}{1 + A_i^n} E_i^{n+1}. \end{aligned} \quad (40)$$

Step 3: Expand the diffusion stencil and collect E^{n+1} terms. Expand the diffusion operator:

$$\begin{aligned} & -\frac{1}{\Delta z} \left(D_{i+\frac{1}{2}}^n \frac{E_{i+1}^{n+1} - E_i^{n+1}}{\Delta z} - D_{i-\frac{1}{2}}^n \frac{E_i^{n+1} - E_{i-1}^{n+1}}{\Delta z} \right) \\ &= -\frac{1}{\Delta z^2} \left(D_{i+\frac{1}{2}}^n (E_{i+1}^{n+1} - E_i^{n+1}) - D_{i-\frac{1}{2}}^n (E_i^{n+1} - E_{i-1}^{n+1}) \right) \\ &= -\frac{D_{i+\frac{1}{2}}^n}{\Delta z^2} E_{i+1}^{n+1} + \frac{D_{i+\frac{1}{2}}^n}{\Delta z^2} E_i^{n+1} + \frac{D_{i-\frac{1}{2}}^n}{\Delta z^2} E_i^{n+1} - \frac{D_{i-\frac{1}{2}}^n}{\Delta z^2} E_{i-1}^{n+1} \\ &= -\frac{D_{i-\frac{1}{2}}^n}{\Delta z^2} E_{i-1}^{n+1} + \frac{D_{i+\frac{1}{2}}^n + D_{i-\frac{1}{2}}^n}{\Delta z^2} E_i^{n+1} - \frac{D_{i+\frac{1}{2}}^n}{\Delta z^2} E_{i+1}^{n+1}. \end{aligned} \quad (41)$$

Step 4: Move all E^{n+1} terms to the left and identify the coefficients.

Start from the full equation, substitute the simplified RHS (Step 2) and the expanded diffusion term (Step 3):

$$\begin{aligned} \frac{E_i^{n+1} - E_i^n}{\Delta t} - \frac{D_{i-\frac{1}{2}}^n}{\Delta z^2} E_{i-1}^{n+1} + \frac{D_{i+\frac{1}{2}}^n + D_{i-\frac{1}{2}}^n}{\Delta z^2} E_i^{n+1} - \frac{D_{i+\frac{1}{2}}^n}{\Delta z^2} E_{i+1}^{n+1} \\ = \frac{\chi c \sigma_i^n}{1 + A_i^n} U_{R_i}^n - \frac{\chi c \sigma_i^n}{1 + A_i^n} E_i^{n+1}. \end{aligned} \quad (42)$$

Bring the $-\frac{\chi c \sigma_i^n}{1 + A_i^n} E_i^{n+1}$ term to the left, and also move $-\frac{E_i^n}{\Delta t}$ to the right:

$$-\frac{D_{i-\frac{1}{2}}^n}{\Delta z^2} E_{i-1}^{n+1} + \left(\frac{1}{\Delta t} + \frac{D_{i+\frac{1}{2}}^n + D_{i-\frac{1}{2}}^n}{\Delta z^2} + \frac{\chi c \sigma_i^n}{1 + A_i^n} \right) E_i^{n+1} - \frac{D_{i+\frac{1}{2}}^n}{\Delta z^2} E_{i+1}^{n+1} = \frac{E_i^n}{\Delta t} + \frac{\chi c \sigma_i^n}{1 + A_i^n} U_{R_i}^n \quad (43)$$

Step 5: Read off the tridiagonal coefficients. Therefore, the equation has the standard tridiagonal form

$$a_i^n E_{i-1}^{n+1} + b_i^n E_i^{n+1} + c_i^n E_{i+1}^{n+1} = d_i^n, \quad (44)$$

with

$$\boxed{a_i^n = -\frac{D_{i-\frac{1}{2}}^n}{\Delta z^2}, \quad b_i^n = \frac{1}{\Delta t} + \frac{D_{i+\frac{1}{2}}^n + D_{i-\frac{1}{2}}^n}{\Delta z^2} + \frac{\chi c \sigma_i^n}{1 + A_i^n}, \quad c_i^n = -\frac{D_{i+\frac{1}{2}}^n}{\Delta z^2}} \quad (45)$$

and

$$\boxed{d_i^n = \frac{E_i^n}{\Delta t} + \frac{\chi c \sigma_i^n}{1 + A_i^n} U_{R_i}^n.} \quad (46)$$

And from the file `1D Diffusion self similar in gold.py`, the implementation of the implicit step is:

6 Comparing the Numerical results

There were 2 main tests done to verify the code, one for the linear case ($C_V \sim T^3$) and one for the non-linear case (self-similar in gold - comparing to Shussman and Heizler).

6.1 Linear Case

The linear case was tested against the semi-analytic solution provided in the file `1D Diffusion Linear.tex`. In order to compare the self-similar code to the linear code, we had to set the right values for the self-similar parameters:

$$\alpha = 0, \quad \gamma = 4, \quad \lambda = -1, \quad \mu = 1, \quad g = 1, \quad f = \frac{a}{T_0^4}. \quad (47)$$

Where T_0 is the transform of the temperature in Hev to Kelvin, and equals 1.16045×10^6 . At the beginnig we forgot to set $\chi = 1$ in the self-similar code, and the results were off by a lot (as seen in figure 1). The disagreement is most pronounced at early times, because at late times a large χ is closer to the physical regime (strong matter–radiation coupling). We then set $\chi = 1$ in the self-similar code, and the results matched perfectly, as seen in figure 2.

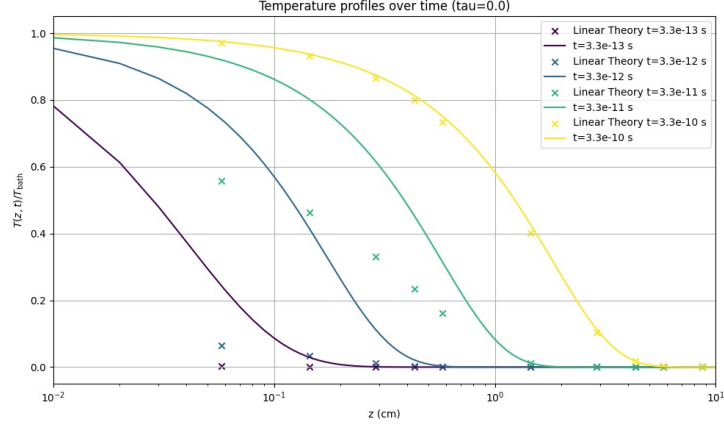


Figure 1: comparison of the 1D diffusion linear code to the self-similar code where $\chi = 10^4$ in the self-similar case.

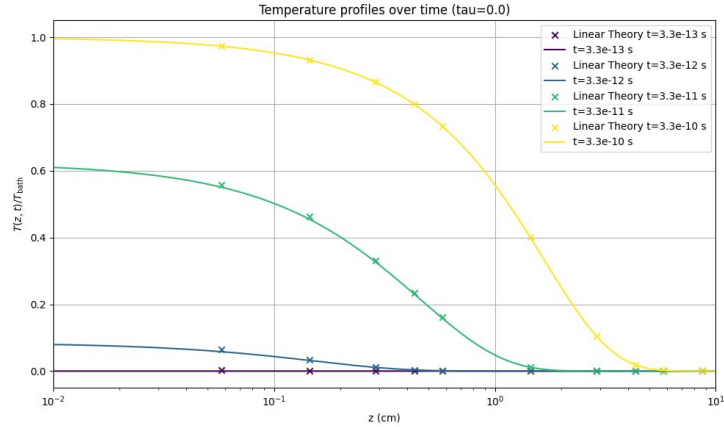


Figure 2: comparison of the 1D diffusion linear code to the self-similar code where $\chi = 1$ in the self-similar case.

6.2 Self-Similar Case in Gold

iiiiiii HEAD The self-similar case in gold was tested against the semi-analytic solution provided in the Shussman and Heizler paper. The comparisons were done for 2 different values of τ : $\tau = 0$ and $\tau = 0.1408$. That because the Shussman and Heizler paper provided results for these 2 values of τ , where $\tau = 0$ is constant temperature boundary condition, and $\tau = 0.1408$ is constant flux boundary condition. The comparisons were done for the front position and for the total energy in the material, where they were calculated in the following

way:

$$z_f(t) = z \quad \text{where} \quad \frac{\partial T(z_f, t)}{\partial z} \text{ is maximal} \quad (48)$$

$$E_{total}(t) = \int_0^{z_F(t)} U_m(z, t) dz \quad (49)$$

The equations for the front position and total energy were taken from the Shussman and Heizler paper. For $\tau = 0$:

$$z_f(t) = 11.53 \cdot 10^{-4} \rho_0^{-1.03} T^{1.95} t^{0.5} \quad (50)$$

$$E_{total}(t) = 0.29 \rho_0^{-0.17} T^{3.55} t^{0.5} \quad (51)$$

And for $\tau = 0.1408$:

$$z_f(t) = 8.79 \cdot 10^{-4} \rho_0^{-1.03} T^{1.95} t^{0.775} \quad (52)$$

$$E_{total}(t) = 0.21 \rho_0^{-0.17} T^{3.55} t \quad (53)$$

For $N_z = 1000$ and a changing time step ($dt_0 = 5 \cdot 10^{-15}$ and $dt_{max} = 5 \cdot 10^{-13}$), the results matched very well, as seen in figures 3, 4, 5 and 6, for $N_z = 10000$ the results were even better (not shown). ===== The self-similar case in gold was tested against the semi-analytic solution provided in the Shussman and Heizler paper. `lllllll d1cae19035fa5e340e10bfd1a28016377da5c2c`

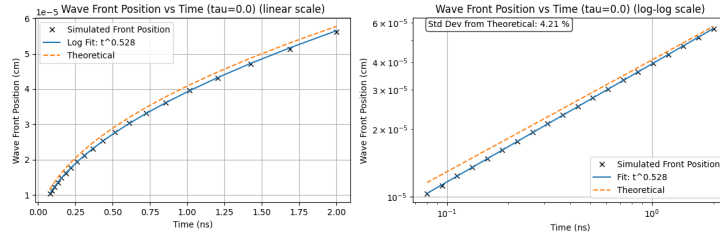


Figure 3: comparison of the front position ($\tau = 0$) of the heat wave from 1D diffusion self-similar code to the Shussman and Heizler semi-analytic solution in gold.

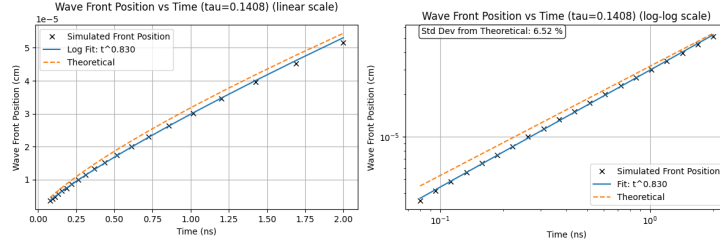


Figure 4: comparison of the front position ($\tau = 0.1408$) of the heat wave from 1D diffusion self-similar code to the Shussman and Heizler semi-analytic solution in gold.

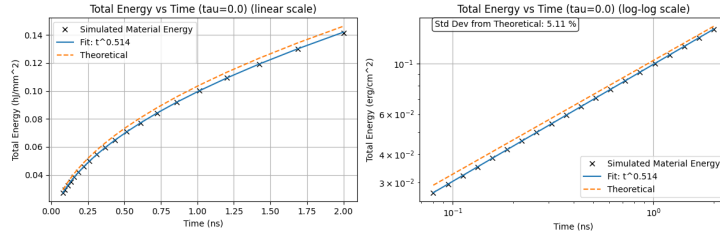


Figure 5: Comparison of the total energy ($\tau = 0$) from 1D diffusion self-similar code to the Shussman and Heizler semi-analytic solution in gold.

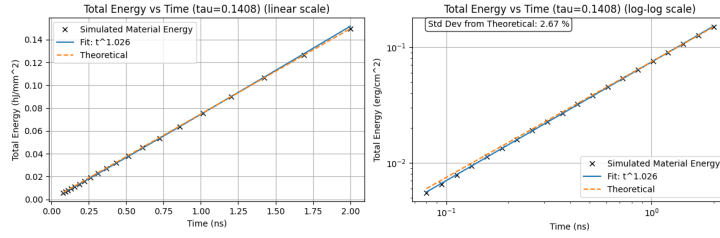


Figure 6: Comparison of the total energy ($\tau = 0.1408$) from 1D diffusion self-similar code to the Shussman and Heizler semi-analytic solution in gold.